TEST PLAN DOCUMENT

- Project Name - TutorLink
- Document Name- Test Plan
- Version-1.0
- Prepared By-Tishany T. (CT/2021/088)
- Role-Developer & QA Tester
- Date-24 December 2025
- Application Type-Web Application

1. Objective

The objective of this test plan is to define the testing strategy, scope, approach, resources, and schedule for the TutorLink web application. This document aims to ensure that all functionalities of TutorLink operate correctly according to the specified requirements and provide a reliable and user-friendly experience for both students and tutors.

2. Project Overview

TutorLink is a web-based platform developed to improve communication between university students and tutors. The system allows students to sign up, log in, view available tutors, send tutoring requests, track the status of requests, and submit feedback after sessions. Tutors log in using pre-created accounts, view and manage student requests, accept or reject tutoring sessions, add common notices, and view student feedback. The platform uses Firebase Authentication for secure login and Cloud Firestore for real-time data storage.

3. Scope of Testing

3.1 In-Scope

The following functionalities are included in testing:

- Home page access
- Student signup
- Student login and authentication
- Student dashboard
- Viewing available tutors
- Sending tutoring requests
- Viewing request status
- Submitting feedback
- Viewing common notices
- Updating student profile
- Tutor login (pre-created accounts)

- Tutor dashboard
- Accepting tutoring requests
- Rejecting tutoring requests
- Adding common notices
- Viewing feedback
- Logout functionality for both users

3.2 Out-of-Scope

The following are excluded from this testing cycle:

- Performance testing
- Load and stress testing
- Security penetration testing
- Mobile application testing
- Third-party integration testing

4. Testing Approach

Testing will be performed manually by executing predefined test cases derived from the use-case diagram and system requirements. Each functionality will be verified against expected results. Defects identified during testing will be documented in a defect report, fixed, and retested to ensure proper resolution.

5. Types of Testing

The following testing types will be conducted:

- Functional Testing – To verify system features function as expected
- UI Testing – To ensure correct layout, navigation, and readability
- Smoke Testing – To confirm basic system stability
- Regression Testing – To verify that fixes do not affect existing features

6. Test Environment

Item	Description
Browsers	Google Chrome , Mozilla Firebox
Operating Systems	Windows 11
Backend Services	Firebase Authentication
Database	Firebase Cloud Firestore
Environment	Localhost / Firebase Hosted Web App

7. Entry Criteria

Testing activities will begin when:

- Application is deployed and accessible
- Functional requirements are finalized
- Use-case diagram is completed
- Test cases are prepared and reviewed

8. Exit Criteria

Testing will be considered complete when:

- All planned test cases have been executed
- All critical and high-severity defects are resolved
- Test summary and approval documents are prepared

9. Roles and Responsibilities

Role: Developer & QA Tester

Name: Tishany T. (CT/2021/088)

Responsibilities:

- Designed and developed the TutorLink web application
- Prepared test plan, test scenarios, and test cases
- Executed manual functional and UI testing
- Identified, documented, and fixed defects
- Performed retesting and regression testing
- Prepared test summary and testing approval documents

10. Test Deliverables

The following documents will be produced during the testing process:

- Test Plan Document
- Test Scenario Document
- Test Case Document
- Test Execution Report
- Defect Report
- Test Summary Report
- Test Approval / Sign-off Document

11. Risks and Mitigation

Risk	Mitigation
Firebase authentication issues	Retest after configuration fixes
Role-based access problems	Validate permissions for student and tutor roles
Real-time data update failures	Retest Firestore listeners

12. Test Schedule

Activity	Duration
Test Planning	1 Day
Test Case Preparation	2 Days
Test Execution	3 Days
Defect Fixing & Retesting	2 Days
Final Reporting	1 Day

13. Approval

Tested and Approved By:

Name: Tishany T.

Role: Developer & QA Tester

Signature: Tishany T.

Date: 24 December 2025